



BITS Pilani
DIGITAL

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 5

Equity Portfolio Management (Part 1)



STUDY NOTES • EDITION 1.0 • WEEK 5

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How to use these notes



These notes cover the whole of Week 5: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A defines return and risk and shows how to compute both from a column of prices. Part B combines assets, which is where diversification stops being a slogan and becomes a calculation. Part C prices the risk that survives diversification. Read the parts in order the first time; each assumes the last, and the same two stocks run through all nine topics.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

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5	The two-asset portfolio
6	The efficient frontier

C Pricing risk

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CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

Module overview



This module is where the course turns quantitative. It defines the two questions every investment raises — return and risk — shows how to compute both from freely available price data, then demonstrates that combining imperfectly correlated assets produces a portfolio less risky than the average of its parts. It closes by pricing the risk that diversification cannot remove.

One thread runs through all nine topics: **you are paid for the risk you cannot diversify away**. Everything before CAPM establishes what that risk is; CAPM puts a number on what it should earn.

Learning objectives

By the end of this module, you will be able to:

- 1 **Define** return and risk, compute the mean and standard deviation of a return series, and apply the rational investor rule.
- 2 **Explain** correlation and compute the return and risk of a two-asset portfolio.
- 3 **Describe** the efficient frontier, identify dominated portfolios, and explain the market portfolio.
- 4 **Apply** the Capital Asset Pricing Model and interpret beta and the security market line.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	What a portfolio is, and the two questions	A • Single asset	L14 V1	LO 1
2	Measuring risk: sigma and the bell curve	A • Single asset	L14 V1	LO 1
3	From prices to two numbers	A • Single asset	L15 V1	LO 1
4	Correlation: why imperfect stocks help	B • Combining	L17 V1	LO 2
5	The two-asset portfolio	B • Combining	L17 V1	LO 2
6	The efficient frontier	B • Combining	L18 V1	LO 3
7	From sigma to beta	C • Pricing risk	L18–L19 V1	LO 3, 4
8	The Capital Asset Pricing Model	C • Pricing risk	L19 V1	LO 4
9	Reading beta	C • Pricing risk	L19 V1	LO 4

Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Portfolio

A curated collection of assets assembled for an objective.

Risk

The uncertainty around whether the expected reward will arrive.

Time 0 and time 1

The present, where the decision is made, and the future, where the outcome is evaluated.

Systematic risk

Market-wide risk that diversification cannot remove.

Correlation (ρ)

The degree to which two assets move together, ranging from -1 to $+1$.

Weights

The proportions allocated to each asset, summing to 1 and never negative.

Minimum variance portfolio

The leftmost point on the curve — the lowest risk achievable from the given assets.

Market portfolio

All risky assets weighted by market capitalisation; proxied by an index fund.

Beta (β)

A stock's systematic risk relative to the market; the slope of its regression on market returns.

Market risk premium

The excess of expected market return over the risk-free rate; historically 5–8%.

CAPM

The Capital Asset Pricing Model, linking required return to systematic risk.

Return

The reward from an investment, measured as percentage change in price.

Sigma (σ)

Standard deviation — how widely returns spread around their average; the single number for risk.

Company-specific risk

Risk belonging to one firm; also called diversifiable, unsystematic, unique or idiosyncratic.

Diversification

Holding many assets so that individual shocks partly cancel.

Asset allocation

The decision of how much money to place in each asset.

Efficient frontier

The upper edge of the risk-return curve, where no portfolio is dominated.

Dominated portfolio

One for which another offers more return at the same risk.

Passive investing

Holding the market portfolio rather than selecting individual stocks.

Risk-free rate

The baseline return on a completely safe investment; in India, the T-bill or G-Sec yield.

Security market line

The plot of expected return against beta; intercept R_f , slope the market risk premium.

PART A • RETURN, RISK AND THE SINGLE ASSET

What return and risk actually are

1 What a Portfolio Is, and the Two Questions

PART A • RETURN, RISK AND THE SINGLE ASSET • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Every investment raises exactly two questions — how much do you expect to gain, and how certain is it — and the whole of portfolio theory is the attempt to answer the second one numerically.

Why this matters

This week is where the course turns quantitative. Everything from here to Week 8 rests on the return-and-risk pair defined in this topic, so it is worth being precise rather than approximately right.

A portfolio is a selection, not a pile

A photographer selects her finest images into one folder. An architect gathers his best designs to win clients. In each case the collection is *curated* for an objective. In finance the items are financial assets, and a collection consisting entirely of stocks is an **equity portfolio**.

The two questions

Return is the reward. How much do you expect to gain? Buy shares of Infosys and the return is the reward you are reaching for. **Risk is the uncertainty.** Will that reward actually arrive, or could things go the other way? Risk is the doubt that shadows every reward.

A sensible investor weighs expected reward *and* the uncertainty around it — never reward alone. That is the entire discipline in one sentence.

The investment horizon

Every financial decision is reasoned between two moments. At **time 0**, the present, you analyse the opportunity and commit your money; every decision is made here. At **time 1**, the future, you evaluate what actually happened — the realised return and the risk you carried.

The horizon may be minutes for a trader or decades for a retirement saver. It stretches or shrinks, but the logic never changes: reason forward from time 0 to time 1.

Company-specific risk, and the escape from it

Four names describe one idea: *diversifiable*, *unsystematic*, *unique*, *idiosyncratic*. These are risks belonging to a single company — a scandal, a factory fire, a lawsuit, a failed product. Enron was erased by accounting fraud. Satyam Computers destroyed shareholder value through fabricated books. Volkswagen was hammered by its emissions scandal.

Each shock struck one firm, not the whole market. And because the risk belongs to one firm, there is a way to escape it: hold many stocks. When one stumbles, another may rise, and the shocks partly cancel. That insight was powerful enough to earn a Nobel Prize.

TWO QUESTIONS, ASKED AT TIME 0 AND ANSWERED AT TIME 1

TIME 0 • THE PRESENT

You analyse the opportunity and commit your money • every decision is made here

THE HOLDING PERIOD

Minutes for a trader • decades for a retirement saver

TIME 1 • THE FUTURE

You evaluate the realised return and the risk actually carried

Return is the reward. Risk is the uncertainty. A sensible investor never weighs one without the other — and the whole of this week is about measuring the second.

Figure 1.1 The investment horizon. The length changes; the logic never does.

KEY IDEA

Company-specific risk is the only kind you can escape without giving up return. That single asymmetry is why diversification is not merely prudent but *free* — and why it earned a Nobel Prize rather than a footnote.

EXAM POINTER

Give all four names for company-specific risk — diversifiable, unsystematic, unique, idiosyncratic — since questions may use any of them. Then name at least two real examples and say explicitly that each shock struck one firm rather than the market. That distinction is what makes diversification work.

QUICK RECAP • TOPIC 1

- A portfolio is a curated collection of assets assembled for an objective.
- Every investment raises two questions: return as reward, risk as uncertainty.
- Decisions are made at time 0 and evaluated at time 1; only the horizon length varies.
- Company-specific risk has four names and belongs to a single firm — Enron, Satyam, Volkswagen.
- Because it belongs to one firm, holding many stocks escapes it.

2 Measuring Risk: Sigma and the Bell Curve

PART A • RETURN, RISK AND THE SINGLE ASSET • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Risk is the width of the range of things that could happen — and sigma is the single number that measures that width.

Why this matters

Sigma is the workhorse of the next four weeks. It appears in the portfolio risk formula in Topic 5, it defines the efficient frontier in Topic 6, and it is the denominator of the Sharpe ratio in Week 6.

Visualising risk

Picture the range of returns an asset might deliver by time 1. The wider the spread of possible outcomes, the riskier the asset.

Width is risk made visible.

A high-risk asset might range from -90% to +200% — a wide, flat bell curve where almost anything could happen. A low-risk asset might range from -10% to +10% — a narrow, tall curve where outcomes huddle together.

Sigma

Sigma (σ) is standard deviation. It measures how far returns stray from their average, shares the unit of what it measures, and serves as our single number for risk. A bigger sigma means a bumpier, riskier ride. On the bell curve, sigma measures the width.

The rational investor rule

Faced with two assets offering the same expected return — say 12% each — the rational investor always chooses the one with the lower sigma. Why accept more uncertainty for the very same reward?

This rule looks trivial. It is not: it is the criterion that will later separate the efficient half of the frontier from the dominated half, and it is what makes one portfolio objectively better than another rather than merely different.

SAME REWARD, DIFFERENT WIDTH • THE RATIONAL INVESTOR ALWAYS TAKES THE NARROWER CURVE

HIGH-RISK ASSET
RANGE OF RETURNS

-90% to +200%

BELL CURVE SHAPE

Wide and flat

SIGMA

Large — uncertain

OUTCOME SPREAD

Almost anything could happen

INVESTOR APPEAL

Only if the reward is higher

LOW-RISK ASSET
RANGE OF RETURNS

-10% to +10%

BELL CURVE SHAPE

Narrow and tall

SIGMA

Small — predictable

OUTCOME SPREAD

Outcomes huddle together

INVESTOR APPEAL

Preferred at equal reward

*At **equal expected return**, lower sigma always wins. That is the rule that makes one portfolio objectively better than another rather than merely different.*

Figure 2.1 Risk as width. Sigma is the number that measures it.

KEY IDEA

Sigma is not a measure of how badly things might go. It measures how *widely* they might go, in either direction. An asset with a large sigma is as likely to surprise you pleasantly as unpleasantly — which is precisely why it must offer more to be worth holding.

COMMON PITFALL

Treating a high sigma as automatically bad. It is a description, not a verdict. A high-sigma asset is a perfectly rational holding provided the expected return compensates — and Topic 5 shows it can even reduce total portfolio risk when combined well.

QUICK RECAP • TOPIC 2

- Risk is the width of the range of possible returns; the bell curve makes it visible.
- Sigma is standard deviation — the single number measuring that width.
- A wide flat curve means large sigma; a narrow tall curve means small sigma.
- At equal expected return, the rational investor always chooses lower sigma.
- Sigma describes dispersion in both directions, not just downside.

3 From Prices to Two Numbers

PART A • RETURN, RISK AND THE SINGLE ASSET • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

A column of free, public closing prices yields exactly two numbers: the reward and the risk.

Why this matters

This is the practical pipeline every later calculation depends on. Get the mechanics right — particularly the n minus one rule — and everything downstream follows.

Why we watch the price

Corporate data — statements, annual reports, board minutes — is slow, backward-looking, uneven, and can be faked; Satyam showed that reports can lie. Market data is one number that moves live, prices in everything, and reacts immediately.

Price already absorbs the judgement of every investor, and it arrives on your screen from the same place it reaches everyone else: exchanges such as the NSE and BSE, aggregators such as Yahoo and Screener, and brokerages such as Zerodha and Groww. Since everyone has the same raw material, it is **analysis, not access**, that separates winners from losers.

Step one: price into return

$R = (P_1 - P_0) \div P_0$ — today's price minus yesterday's, divided by yesterday's.

Take ABC Ltd. The 1 May close is ₹100 and the 2 May close is ₹98.30, so $R = (98.30 - 100) \div 100 = -1.73\%$. Day one has no return, because there is no price before it. **Fifteen prices therefore give only fourteen returns** — n prices yield $n - 1$ returns, a rule worth remembering.

Steps two and three: the mean and sigma

Add all fourteen returns and divide by fourteen. For ABC Ltd this gives an average daily return of **-1.23%**. On a typical day in this window the stock lost just over one per cent. That is the reward side — but a stock that loses one per cent on a calm day is very different from one that swings wildly to get there.

Sigma supplies the other half. $\sigma = \sqrt{[\sum(R_i - \text{mean})^2 \div (n - 1)]}$, where $n = 14$. We divide by $n - 1$, not n , because we are working with a sample rather than the whole population. In Excel the shortcut is **=STDEV(range)**, and for ABC Ltd it returns **1.66%** — a daily figure, in the same unit as the daily returns it measures.

ONE COLUMN OF PRICES • TWO NUMBERS OUT

15 CLOSING PRICES

Free and public • from the NSE, an aggregator or a broker

14 DAILY RETURNS

$R = (P_1 - P_0) \div P_0$ • n prices give $n - 1$ returns

MEAN = -1.23%

The reward •
=AVERAGE(returns)

SIGMA = 1.66%

The risk • =STDEV(returns)

Both figures are **daily**, computed from nothing but a column of free public prices. The example treats all fifteen days as trading days, though real markets close at weekends.

Figure 3.1 The measurement pipeline for ABC Ltd. Every later calculation starts here.

COMMON PITFALL

Running AVERAGE and STDEV on the *prices* rather than the returns. Prices trend, returns do not, and the standard deviation of a price series measures something meaningless. Convert first, always.

EXAM POINTER

State the return formula, then note the $n - 1$ rule for both the count of returns and the sigma denominator — both are commonly tested. When asked to compute, show the conversion step explicitly rather than jumping to the answer.

QUICK RECAP • TOPIC 3

- Market prices are live, public and absorb every investor's judgement; analysis, not access, differentiates.
- $R = (P_1 - P_0) \div P_0$. Fifteen prices give fourteen returns.
- The mean of those returns is the reward: -1.23% daily for ABC Ltd.
- $\sigma = \sqrt{[\sum(R_i - \text{mean})^2 \div (n - 1)]}$; in Excel, =STDEV(range). For ABC Ltd, 1.66% .
- Divide by $n - 1$ because the data is a sample, not the whole population.

PART B • COMBINING ASSETS

Where diversification becomes a calculation

4 Correlation: Why Imperfect Stocks Help

PART B • COMBINING ASSETS • TOPIC 4 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

The secret to lower risk is not finding one flawless stock. It is holding two imperfect ones that do not move in step.

Why this matters

Correlation is the mechanism behind every diversification claim in this course. Without it, portfolio risk would simply be a weighted average and none of the rest of the week would exist.

The counterintuitive core

When two stocks do not move together, a day that pulls one down may push the other up, and their swings partly cancel. When tech sentiment sours, energy may hold firm — an Infosys and Reliance holding rides steadier than either name alone.

Note carefully what has *not* happened: neither stock became less volatile. The offsetting movements produce a smoother path for the combination without removing either stock's individual volatility.

Correlation, measured

Correlation measures the degree to which two stocks move together, ranging from -1 to $+1$.

At $\rho = +1$ the assets move perfectly together, rising and falling as one, and there is no diversification benefit whatsoever. At $\rho = 0$ there is no consistent pattern — one stock's movement gives no clue about the other — and the benefit is moderate. At $\rho = -1$ the assets move in exactly opposite directions and the benefit is at its maximum.

In Excel: `=CORREL(series1, series2)`. Always use **returns**, never prices. For ABC and XYZ the correlation is **0.33**.

Weights: the one lever you control

Each stock's return, each stock's risk and the correlation between them are all fixed by the market. Exactly one decision is left to you: the proportion of money placed in each asset. This is **asset allocation**, and the proportions are called **weights**.

With ₹10,000 of capital, ₹3,000 in ABC gives a weight of 0.3 and ₹7,000 in XYZ gives 0.7. The weights sum to 1.0, so every rupee is accounted for, and neither may be negative.

CORRELATION RUNS FROM -1 TO $+1$ • LOWER IS BETTER

Correlation	What it means	Diversification benefit
$\rho = +1$	Assets move perfectly together — they rise and fall as one	None
$\rho = 0$	No consistent pattern; one gives no clue about the other	Moderate
$\rho = -1$	Assets move in exactly opposite directions	Maximum
ABC and XYZ	Measured at $\rho = 0.33$ — they drift the same way, but weakly	Substantial, as Topic 5 shows

Two friends walking: if one slows whenever the other speeds up, their **combined pace is steadier** than either alone. Stocks behave the same way.

Figure 4.1 Correlation and the diversification it permits. Use =CORREL on returns, never on prices.

KEY IDEA

Wealth is created by **intelligently combining imperfect stocks**, not by finding one perfect one. The search for the single best stock is the wrong search.

EXAM POINTER

State the -1 to $+1$ range and what each endpoint means for diversification. The mark most often lost is failing to say that CORREL must be run on returns rather than prices — an error that produces a plausible number and a wrong answer.

QUICK RECAP • TOPIC 4

- Combining imperfectly correlated stocks smooths the return path without reducing either stock's volatility.
- Correlation ranges from -1 to $+1$: $+1$ gives no benefit, -1 gives maximum benefit.
- In Excel, =CORREL(returns A, returns B) — returns, never prices.
- ABC and XYZ have a correlation of 0.33.
- Returns, risks and correlation are fixed by the market; only the weights are yours to choose.

5 The Two-Asset Portfolio

PART B • COMBINING ASSETS • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Portfolio return is a simple weighted average. Portfolio risk is not — and the gap between the two is the whole value of diversification.

Why this matters

This is the single most important calculation in the week, and the one most likely to appear as a numerical question. The asymmetry it demonstrates is the foundation of the efficient frontier in Topic 6.

The two inputs

ABC Ltd has a mean daily return of -1.23% and a sigma of 1.66% . XYZ Ltd has a mean of -0.42% and a sigma of 2.07% — the riskier stock. Both are computed over the identical fifteen-day window using AVERAGE and STDEV on the fourteen returns, which is what lets us compare them and measure how they move together.

The two formulas

Portfolio return: $E(R_p) = W_a \times E(R_a) + W_b \times E(R_b)$. The portfolio return is simply the weighted average of the individual asset returns.

Portfolio risk: $\sigma_p = \sqrt{[(W_a^2 \times \sigma_a^2) + (W_b^2 \times \sigma_b^2) + (2 \times \rho_{ab} \times W_a \times W_b \times \sigma_a \times \sigma_b)]}$. Risk is **not** a weighted average — and the third term, containing the correlation, is exactly what allows portfolio risk to fall below one.

Constraints: $W_a + W_b = 1$, and both weights must be zero or greater.

Where the risk goes

Common sense expects a mix of a 1.66% risk and a 2.07% risk to land near their average of 1.86% . It does not. The portfolio's risk comes out *below* that — and we removed none of either stock's volatility. The answer is correlation: because the two do not move in lockstep, their offsetting movements absorb part of the shock.

THE CALCULATION THAT PROVES THE POINT

Step	Working	Result
Inputs	ABC: W 0.3, return -1.23% , σ 1.66% • XYZ: W 0.7, return -0.42% , σ 2.07% • $\rho = 0.33$	—
Portfolio return	$(0.3 \times -1.23) + (0.7 \times -0.42) = -0.369 - 0.294$	-0.66%
Risk, term 1	$0.3^2 \times 1.66^2$	0.248
Risk, term 2	$0.7^2 \times 2.07^2$	2.100
Risk, term 3	$2 \times 0.33 \times 0.3 \times 0.7 \times 1.66 \times 2.07$	0.476
Portfolio risk	$\sqrt{(0.248 + 2.100 + 0.476)} = \sqrt{2.824}$	1.68%

1.68% sits **below the 1.86% simple average** of the two risks, and barely above ABC's own 1.66% — even though 70% of the money sits in the riskier stock. The correlation term has done its quiet work.

Figure 5.1 The two-asset calculation. Return is a weighted average; risk is not.

KEY IDEA

Return is the weighted average. Risk is not. That single asymmetry is the proof that **how** stocks combine matters as much as **what** you hold — and it is the reason the next topic's curve bends.

EXAM POINTER

Numerical questions here are near-certain. Write both formulas, substitute, and show all three terms of the risk calculation separately before taking the square root. State explicitly that the result falls below the simple average and name the correlation term as the reason — that interpretation carries as many marks as the arithmetic.

QUICK RECAP • TOPIC 5

- $E(R_p) = W_a E(R_a) + W_b E(R_b)$ — a simple weighted average.
- σ_p includes a third term containing correlation, which is why it is not a weighted average.
- Weights must sum to 1 and neither may be negative.
- ABC and XYZ at 0.3 and 0.7 give a return of -0.66% and a risk of 1.68%.
- 1.68% is below the 1.86% simple average despite 70% sitting in the riskier stock.

6 The Efficient Frontier

PART B • COMBINING ASSETS • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Vary only the weights and the same two stocks produce hundreds of outcomes — which trace a curve whose upper half is worth holding and whose lower half never is.

Why this matters

Markowitz won a Nobel Prize for this curve. It is also the direct bridge to CAPM: introduce a risk-free asset to the frontier and every investor converges on one portfolio.

The only lever, pushed all the way

Everything is locked except the weights. The returns of ABC and XYZ, their sigmas and the correlation between them are all fixed by the market. Free to move is only the percentage placed in each stock — and pushing that one lever from 100% ABC to 100% XYZ makes the same two stocks yield hundreds of outcomes.

Take eleven sample mixes, labelled A through K. Portfolio A is fully in XYZ; each step shifts 10% across, until portfolio K is fully in ABC. The one rule is that the two weights must always sum to 100%. Portfolio D — 30% ABC, 70% XYZ — is the exact portfolio built in Topic 5, now just one row among eleven.

The curve, and what it tells you

Plot risk on the horizontal axis against return on the vertical and the eleven dots trace a sideways bullet pointing left. As the mix slides across, risk first falls, reaches a low point around portfolios G and H, then climbs again, while return changes steadily across the range.

The **minimum variance portfolio** is the leftmost point — the floor of risk achievable from these two assets — and it splits the curve in two. The **upper half is efficient**: return rises as risk rises, and rational investors choose only from here. The **lower half is dominated**: for any portfolio on it, another sits directly above offering more return at the same risk. Portfolio J sits below E, so nobody picks J.

A note on the negative returns

In this dataset the whole curve sits below zero, because ABC and XYZ happen to have negative average returns over the sample window. This is a data quirk, not a flaw in the method. Rebuild it with two positive-return stocks and the same bullet shape appears, simply floating above the axis. Only the curve's position on the page changes; the logic is identical regardless of sign.

THE FRONTIER • THE MINIMUM VARIANCE PORTFOLIO SPLITS WORTH FROM WORTHLESS

THE EFFICIENT HALF

Above the minimum variance point • return rises with risk • rational investors choose only from here

MINIMUM VARIANCE PORTFOLIO

The leftmost point • the floor of risk • it divides the curve

THE DOMINATED HALF

Below it • another portfolio always offers more return at the same risk • nobody rationally chooses these

Portfolio J sits directly below E — same risk, less return. That is what “dominated” means, and it is the rational investor rule from Topic 2 applied to a whole curve.

Figure 6.1 The efficient frontier. The minimum variance portfolio is the dividing line.

REAL-WORLD EXAMPLE

Harry Markowitz published *Portfolio Selection* in 1952 — just fourteen pages — and was awarded the Nobel Prize in Economics for it in 1990. The central claim is that wealth is not created by hunting for one perfect stock but by combining imperfect ones with skill, because combining assets creates possibilities no single asset holds.

KEY IDEA

Push diversification to its logical end — all risky assets, weighted by market capitalisation — and you reach the **market portfolio**. Unsystematic risk goes to zero and only systematic risk remains. Its real-world proxy is an index fund tracking the Nifty 50 or the Sensex, which is the theoretical foundation of passive investing.

EXAM POINTER

Describe the bullet shape, name the minimum variance portfolio as the leftmost point, and define *dominated* precisely: another portfolio offers more return at the same risk. Adding the 1952 and 1990 dates for Markowitz costs one line and shows command of the material.

QUICK RECAP • TOPIC 6

- Varying only the weights generates a family of portfolios plotting as a leftward bullet.
- The minimum variance portfolio is the leftmost point and the floor of achievable risk.
- The upper half is efficient; the lower half is dominated and never rationally chosen.
- Negative returns in the example shift the curve down the page but change nothing about its logic.
- Extending diversification to all risky assets gives the market portfolio — the basis of index investing.

PART C • PRICING RISK

Putting a price on the risk that remains

7 From Sigma to Beta

PART C • PRICING RISK • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Once you hold the market portfolio, a stock's own wobble is cancelled by others. What remains is beta — and beta is what you are paid for.

Why this matters

This is the conceptual hinge between Markowitz and CAPM. Sigma governed the first two parts of the week; beta governs the third and all of Week 6.

Why the measure has to change

CAPM assumes investors hold well-diversified portfolios. Under that assumption, unsystematic risk — the risk specific to one firm — has been diversified away to almost nothing. What remains is **systematic risk**: the market-wide risk no amount of diversification can remove.

So a stock's sigma is no longer the right gauge. Sigma measures *total* risk, including the part your other holdings already cancel. Beta isolates the part that survives.

What beta measures

Beta measures **sensitivity, not quality**: if the Nifty 50 rises 1%, does this stock rise more, less, or move the opposite way? It says nothing about whether a stock is good, only how aggressively it rides the market's moves.

It is estimated by regressing the stock's returns on the market's returns. In Excel: **=SLOPE(stock returns, market returns)** — stock returns first, market returns second. Swap the order and Excel returns a completely wrong beta. This is the rule that trips everyone up.

The intuition check

Consider two portfolios. Portfolio X returns 1.09% at a risk of 0.10; Portfolio Y returns the same 1.09% at a risk of 2.49. When two portfolios offer identical return, risk is all that matters — X wins, and Y is dominated and inefficient. This is the rational investor rule again, and it is exactly why the market portfolio is special: it offers the best return available for its level of risk.

TWO MEASURES OF RISK • ONLY ONE OF THEM EARNS A REWARD

SIGMA • STANDARD DEVIATION

MEASURES

Total risk

WHAT THAT MEANS

The full swing of a stock on its own

INCLUDES

The wobble that diversification can cancel out

RELEVANCE TO CAPM

Not what you are paid for

BETA

MEASURES

Systematic risk only

WHAT THAT MEANS

How much a stock moves when the market moves

INCLUDES

Only the risk a diversified investor cannot escape

RELEVANCE TO CAPM

The risk CAPM compensates

*Hold the market portfolio and one stock's wobble is cancelled by others. **Beta is what remains, and beta is what you are paid for.***

Figure 7.1 Sigma versus beta. The difference is what diversification has already removed.

COMMON PITFALL

Reversing the SLOPE arguments. The stock is the y variable and the market the x variable, so stock returns come first. Reversing them produces a plausible-looking number that is entirely wrong, and Excel gives no warning.

EXAM POINTER

State the distinction in one sentence — sigma is total risk, beta is systematic risk — then explain *why* the measure changes: because CAPM assumes diversification has already removed the rest. That causal step is what the question is testing.

QUICK RECAP • TOPIC 7

- CAPM assumes diversified investors, so unsystematic risk has already been removed.
- Sigma measures total risk; beta isolates the systematic part that survives diversification.
- Beta measures sensitivity to the market, not quality of the stock.
- In Excel, =SLOPE(stock returns, market returns) — order matters absolutely.
- At equal return, the lower-risk portfolio dominates — which is why the market portfolio is special.

8 The Capital Asset Pricing Model

PART C • PRICING RISK • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

You are not paid for the return you hope for. You are paid for the risk you agree to carry — and CAPM prices that risk exactly.

Why this matters

CAPM is the destination of the entire week and the tool used throughout Weeks 6 and 7. It converts a measure of risk into a required rate of return, which is what makes valuation possible at all.

The oldest bargain in finance

Why accept 7% when 18% is on the table? A fixed deposit at 7% is near certain but modest; you are paid almost nothing extra because you carry almost no risk. Equity at 18% carries a real chance of loss.

The golden rule: **expected return = risk-free rate + risk premium**. The *risk-free rate* is the reward for waiting — pure compensation for time, taken in India from the 91-day treasury bill or the 10-year government security. The *risk premium* is the reward for bearing uncertainty. CAPM makes that premium exact.

From Markowitz to CAPM in seven steps

Weights build portfolios by combining assets in different proportions. Plotting every portfolio on risk against return traces the frontier. Only the upper edge is worth holding. Introduce a risk-free asset, and all investors converge on one portfolio — the market portfolio. Under perfect diversification, unsystematic risk goes to zero. What remains is systematic risk, measured by beta. Expected return therefore becomes proportional to beta.

Markowitz laid the foundation in 1952; Sharpe, Lintner and Mossin completed the bridge in the 1960s. Risk became beta.

The equation

$$E(R_i) = R_f + \beta \times [E(R_m) - R_f]$$

The *market risk premium* is $E(R_m) - R_f$, historically 5 to 8%. The *security risk premium* is beta multiplied by that. Three inputs in, one required return out.

FOUR TERMS, ONE OUTPUT

Term	Name	Meaning	Indian source
R_f	Risk-free rate	The return from a completely safe investment — the baseline	91-day T-bill or 10-year G-Sec yield, from the RBI
$E(R_m)$	Market return	What the overall market is expected to deliver	Historical index averages — the Nifty 50 has delivered about 12–15%
β	Beta	This stock's systematic risk relative to the market	Estimated in Excel with =SLOPE on return series
$E(R_i)$	Required return	The single output — what the investor should demand	Solved for

Expected return = risk-free rate + risk premium. Everything CAPM does is make the second term precise instead of intuitive.

Figure 8.1 The CAPM inputs. Three are supplied or estimated; the fourth is what you solve for.

REAL-WORLD EXAMPLE

Solving for the required return. Given $R_f = 6\%$, $E(R_m) = 15\%$ and $\beta = 1.2$. The market risk premium is $15\% - 6\% = 9\%$. Expected return = $6\% + (1.2 \times 9\%) = 6\% + 10.8\% = \mathbf{16.8\%}$. A beta of 1.2 means 20% more systematic risk than the market, and 10.8% is the exact compensation for it.

EXAM POINTER

Write the equation before substituting, and compute the market risk premium as a separate step — examiners award it separately. Note that the model can be run backwards: given $E(R_i) = 24\%$ with the same R_f and $E(R_m)$, the security risk premium is 18%, the market premium 9%, so $\beta = 18 \div 9 = \mathbf{2.0}$.

QUICK RECAP • TOPIC 8

- You are paid for risk carried, not return hoped for.
- Expected return = risk-free rate + risk premium.
- $E(R_i) = R_f + \beta[E(R_m) - R_f]$, with the market risk premium historically 5–8%.
- $R_f = 6\%$, $E(R_m) = 15\%$, $\beta = 1.2$ gives a required return of 16.8%.
- Run backwards, a required return of 24% on the same inputs implies a beta of 2.0.

9 Reading Beta

PART C • PRICING RISK • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Four regions of the beta scale, and one of them is a rare gem that pays out precisely when everything else is losing value.

Why this matters

The closing topic of the week. Interpreting beta correctly is what turns CAPM from an equation into a portfolio decision, and it sets up the hedging arguments of Week 10.

The four regions

β greater than 1 — aggressive. The stock amplifies market moves in both directions. Small caps, metals and real estate typically sit here.

β equal to 1 — in step. The stock moves exactly with the market. A diversified large cap or an index fund behaves this way.

β between 0 and 1 — defensive. The stock mutes the swings: stable, low drama. Fast-moving consumer goods, pharmaceuticals and utilities are the usual examples.

β below 0 — insurance. The asset moves opposite to the market. Gold ETFs are the standard case, rising when equities crash.

Why negative beta is prized

A negative-beta asset is a rare gem — a natural insurance policy that pays out precisely when everything else in the portfolio is losing value. Few real assets behave this way, which is exactly why they are so valuable, and why serious investors hunt for low or negative correlations rather than merely high returns.

The security market line

Plot the CAPM equation with beta on the horizontal axis and expected return on the vertical, and you get the **security market line**. Its intercept is the risk-free rate, where beta is zero. Its slope is the market risk premium, whose steepness reflects how much extra return each unit of beta earns.

Position tells you pricing. A stock *on* the line is fairly priced, offering exactly the return its risk justifies. *Above* the line it is underpriced — offering more return than its risk warrants. *Below* the line it is overpriced.

FOUR REGIONS OF THE BETA SCALE

Beta	Character	Behaviour	Typical examples
$\beta > 1$	Aggressive	Amplifies market moves in both directions	Small caps, metals, real estate
$\beta = 1$	In step	Moves exactly with the market	A diversified large cap or index fund
$0 < \beta < 1$	Defensive	Mutes the swings — stable, low drama	FMCG, pharmaceuticals, utilities
$\beta < 0$	Insurance	Moves opposite to the market	Gold ETFs, which rise when others crash

On the security market line: **on the line is fairly priced, above it underpriced, below it overpriced.** The intercept is R_f and the slope is the market risk premium.

Figure 9.1 Reading beta. The fourth row is rare, and that rarity is what makes it valuable.

KEY IDEA

A negative beta does not mean a bad investment. It means an asset whose payoff is *timed* to arrive when you need it most — which is a different kind of value from a high expected return, and one a diversified investor will pay for.

EXAM POINTER

Give all four beta regions with a numeric range, a character label and an example each. For the security market line, name the intercept and the slope explicitly, then state the three pricing positions. Questions often ask you to place a stock given its beta and actual return — compute the CAPM return first, then compare.

QUICK RECAP • TOPIC 9

- $\beta > 1$ is aggressive; $\beta = 1$ tracks the market; $0 < \beta < 1$ is defensive; $\beta < 0$ is insurance.
- Negative-beta assets pay out when everything else is falling, which is why they are prized.
- The security market line plots expected return against beta.
- Its intercept is the risk-free rate; its slope is the market risk premium.
- On the line is fairly priced, above it underpriced, below it overpriced.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Define return and risk and compute both from a price series	Topics 1, 2 and 3 · Figures 1.1, 2.1, 3.1
2	Explain correlation and compute two-asset portfolio return and risk	Topics 4 and 5 · Figures 4.1, 5.1
3	Describe the efficient frontier and the market portfolio	Topics 6 and 7 · Figures 6.1, 7.1
4	Apply CAPM and interpret beta and the security market line	Topics 8 and 9 · Figures 8.1, 9.1

The module in six sentences

Every investment raises two questions — the reward expected and the uncertainty around it — and sigma is the single number that measures the second. A column of free public prices yields both: fifteen prices give fourteen returns, whose mean is the reward and whose standard deviation is the risk. Combining two imperfectly correlated stocks produces a portfolio return that is a simple weighted average and a portfolio risk that is not, because a third term containing correlation pulls it below the average of the parts. Varying only the weights traces a bullet-shaped curve whose upper half is efficient and whose lower half is dominated, and extending that logic to all risky assets gives the market portfolio. Once an investor holds it, company-specific wobble has been cancelled and only systematic risk remains, measured by beta rather than sigma. CAPM then prices that risk exactly: required return equals the risk-free rate plus beta times the market risk premium.

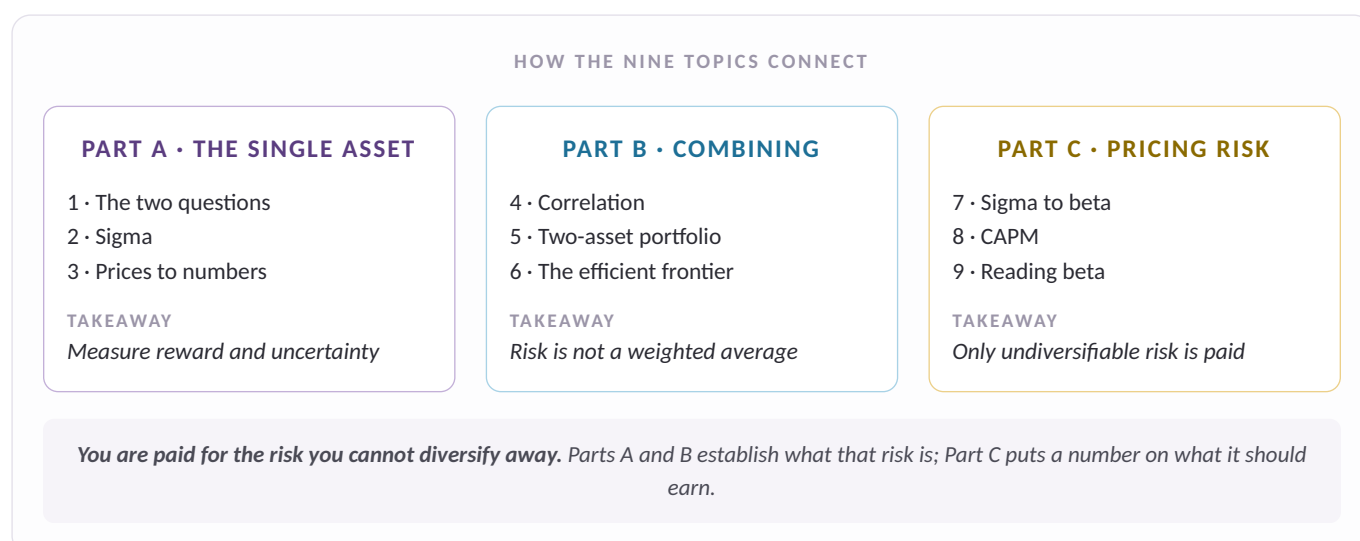


Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check



Term	Definition
Portfolio	A curated collection of assets assembled for an objective.
Return	The reward from an investment, measured as percentage change in price.
Risk	The uncertainty around whether the expected reward will arrive.
Sigma (σ)	Standard deviation — how widely returns spread around their average; the single number for risk.
Time 0 and time 1	The present, where the decision is made, and the future, where the outcome is evaluated.
Company-specific risk	Risk belonging to one firm; also called diversifiable, unsystematic, unique or idiosyncratic.
Systematic risk	Market-wide risk that diversification cannot remove.
Diversification	Holding many assets so that individual shocks partly cancel.
Correlation (ρ)	The degree to which two assets move together, ranging from -1 to $+1$.
Asset allocation	The decision of how much money to place in each asset.
Weights	The proportions allocated to each asset, summing to 1 and never negative.
Efficient frontier	The upper edge of the risk-return curve, where no portfolio is dominated.
Minimum variance portfolio	The leftmost point on the curve — the lowest risk achievable from the given assets.
Dominated portfolio	One for which another offers more return at the same risk.
Market portfolio	All risky assets weighted by market capitalisation; proxied by an index fund.
Passive investing	Holding the market portfolio rather than selecting individual stocks.
Beta (β)	A stock's systematic risk relative to the market; the slope of its regression on market returns.
Risk-free rate	The baseline return on a completely safe investment; in India, the T-bill or G-Sec yield.
Market risk premium	The excess of expected market return over the risk-free rate; historically 5–8%.
Security market line	The plot of expected return against beta; intercept R_f , slope the market risk premium.
CAPM	The Capital Asset Pricing Model, linking required return to systematic risk.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 You have sixty daily closing prices. How many returns can you compute, and what do you divide by when calculating sigma?

Fifty-nine returns, and you divide by fifty-eight. n prices give $n - 1$ returns, because the first day has no prior price. Sigma uses $n - 1$ in the denominator too, since the data is a sample rather than the whole population.

Q2 ABC has a sigma of 1.66% and XYZ 2.07%. Combined at weights of 0.3 and 0.7 the portfolio risk is 1.68%. Why is it below both the simple average and the weighted average?

The correlation term. Portfolio risk includes $2 \times \rho \times W_a \times W_b \times \sigma_a \times \sigma_b$. Because $\rho = 0.33$ rather than 1, the two stocks do not move in lockstep and their offsetting movements absorb part of the risk — even though 70% of the money sits in the riskier stock.

Q3 Portfolio J offers 8% at a risk of 14%. Portfolio E offers 11% at the same 14%. What is J called, and would anyone hold it?

J is dominated, and no rational investor holds it. E offers more return at identical risk. Dominated portfolios lie on the lower half of the curve, below the minimum variance portfolio.

Q4 A stock has a beta of 1.2. The risk-free rate is 6% and the expected market return is 15%. What return should you require, and what does the beta mean?

16.8%. Market risk premium = $15\% - 6\% = 9\%$. Required return = $6\% + (1.2 \times 9\%) = 16.8\%$. A beta of 1.2 means 20% more systematic risk than the market, and the 10.8% premium is the exact compensation for it.

Q5 Why does CAPM use beta rather than sigma?

Because it assumes the investor is already diversified. Sigma measures total risk, including the company-specific part that other holdings have already cancelled. Beta isolates the systematic risk that survives diversification — and only that risk earns a reward.

Q6 A gold ETF has a beta of -0.3. Is that a bad investment?

No — it is a rare and valuable one. A negative beta means the asset rises when the market falls, so it pays out precisely when the rest of the portfolio is losing value. Few real assets behave this way, which is exactly why diversified investors will pay for them.